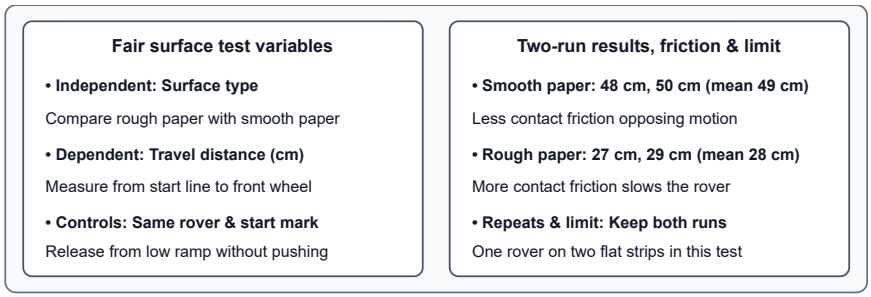
Reference sheets

Reference A · Fair surface test variables and planning rules

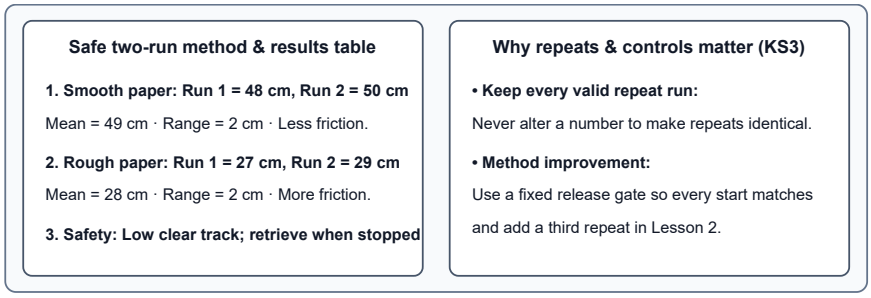


Fair surface test diagram. It shows independent, dependent, and control variables with two-run cm results.

Reference only. Do not write on this sheet. Check each variable role when building your surface plan.

Variable role.	What we do in our test.	Key Stage 3 science reason.	What happens if not controlled.
1. Independent variable (Change).	Change the surface type: rough paper vs smooth paper.	Tests how surface roughness changes contact friction.	If we do not change the surface, we cannot compare friction.
2. Control variables (Keep the same).	Keep the same rover, start mark, and no-push ramp release.	Ensures only surface friction causes any distance change.	A harder push or steeper ramp confounds the result.
3. Dependent variable (Measure).	Measure travel distance in centimetres (cm) from start line to front wheel.	Gives a clear number and unit for every run.	Without cm units, distances cannot be compared fairly.
4. Repeats and test limit.	Complete two repeat runs per surface and keep every valid number.	Shows whether the pattern holds across runs in this test.	One run alone cannot show if results vary.

Reference B · Two-run surface table and safe method



Two-run surface table diagram. It shows smooth and rough cm runs, safety rules, and test limits.

Surface condition.	Run 1 (cm).	Run 2 (cm).	Mean and Key Stage 3 explanation.
Smooth paper.	48 cm.	50 cm.	Mean = 49 cm (range 2 cm). Less contact friction opposes the rover.
Rough paper.	27 cm.	29 cm.	Mean = 28 cm (range 2 cm). More contact friction slows the rover.
Safety and method upgrade.	Low clear track.	Retrieve when stopped.	Keep hands and faces clear. Use a release gate to keep starts identical.

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival diagram. Identify the independent variable, dependent variable, and two control variables.

Help. Match each variable type to Boxes 1, 2, and 3.

2. Why does changing both the surface and the launch push make an investigation invalid?

Help. Explain why only one independent variable is changed.

3. In Box 3, rough paper runs are 27 cm and 29 cm. Calculate the mean and range.

Help. Add both runs and divide by 2 for the mean.

4. State one limitation of testing one small rover across two flat paper strips.

Help. Think about what a small desk setup cannot show.

You do tasks

1. Use Reference A and Reference B. Write a Key Stage 3 fair-test plan naming the independent, dependent, and control variables. Explain how pushing by hand confounds the data. **Staff I ☐ S ☐ C ☐**

Help. Explain how an uncontrolled push mixes launch force with surface friction.

2. Use Reference B (Rough: 27, 29 cm; Smooth: 48, 50 cm). Compare the means and ranges, and explain the pattern using contact friction. **Staff I ☐ S ☐ C ☐**

Help. Find the mean and range for each surface.

3. Two repeat runs on rough paper differ by 2 cm. Explain why we keep both values. State one limitation and one method upgrade. **Staff I ☐ S ☐ C ☐**

Help. Link repeat variation, sample size, and a release gate.

Optional participation-loop prompt. Share your test plan and result with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Why does changing both the surface and the rover break a fair test?

2. Use the means (49 cm vs 28 cm) and contact friction. State what the two-run table shows.

3. State one limitation of this two-run desk test. State one improvement for next lesson.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt